Team meeting briefing: decisions, novelty, technical basis and Thai conditions

Leaf-wrapped frozen coconut-pandan drink pop, Innovative Bev & Bite entry. Prepared 18 October 2026. Virtual evidence only: no physical or consumer test has been run, and none is in scope.

1. Decisions taken (18 October 2026)

• **Working design (confirmed by the user, 18 October):** the leaf-wrapped frozen coconut-pandan drink with a food-grade inner barrier, under Alternative Formats. Leaf-only (no barrier) is rejected on physical grounds: freeze damage and stomata would leak (rework/30, rework/31).

• **Inner barrier:** the inner film is plastic, so the product is not plastic-free. It is the working design, not a fallback.

• **Claim:** describe the product as **leaf-wrapped**. Do not say plastic-free. The outer wrap is leaf; the inner barrier is plastic.

• **Novelty claim:** a leaf-wrapped frozen drink eaten peel-and-sip as it thaws. No such drink was found in the extended search. One leaf-wrapped frozen ice-cream bar is a disclosed precedent (section 3).

• **Hot-season use conditions:** keep the pop in an insulated box with gel packs, keep it out of ice water, and eat it within about an hour when it is in full sun. These go in the product instructions. They do not go on the label until a Thai registration adviser confirms the classification.

• **Product classification for holding rules:** treat the product as a non-heat ready-to-eat beverage. Thai rules then require below 5 °C, or no more than 2 hours at room temperature, when sold from food premises.

• **After thaw (spoilage):** eat within 2 hours of leaving the freezer, and within about 1 hour in full sun. Do not refreeze. Keep the wrap dry and out of standing water. Spoilage needs about 6 to 12 h at 30 C (rework/28), so the Thai 2-hour limit is the operative one. Pathogens: one contaminating cell reaches the hazard in about 6 to 27 h at 30 C, but a serving that starts at the hazard level is unsafe at once. Production control (a validated kill step or pathogen-free raw material) is therefore required (rework/29).

• **Physical test (approved 18 October):** a dye-penetration test on leaf-only wraps (rework/33), with 300 wraps and a pre-registered pass rule. The leaf-only option is approved for testing; the working design stays the barrier until the result is in. The test has not yet been run.

2. The product

A frozen coconut or pandan drink sealed in a banana or pandan leaf. The consumer peels the leaf open and sips the drink as it thaws. The newness is the consumption act, not an ingredient. Target: Thai buyers who value traditional packaging (hypothesis; the evidence is produce wrapping, not drinks). Assumed size: 38 mm core, 41 mm leaf wrap, 150 mm tall, 170 mL.

3. Novelty: searched, one precedent found

• **Searched:** about 137 queries in four searches, covering 11 languages and the web, patents, the OTOP registry, and Thai retail and delivery sites. Matches were re-opened by a separate checker, and each absence claim was tested with new queries.

• **Found:** no leaf-wrapped frozen drink, slush or ice pop. No patent found claims a leaf-based wrap or container for a frozen drink. The only patent near the concept is a pending plant-fibre food wrap, which is not a leaf.

• **Closest precedent:** an old-style ice-cream bar wrapped in banana leaf, sold at Talat O-Ploi market in Suan Phueng, Ratchaburi. It rests on one 2023 user post, and a 2024 review of the same market's bars does not mention the leaf. Its frozen status and whether the leaf is peeled or kept on while eating are inferred, not confirmed.

• **Other leaf items:** a Chanthaburi menu item named "banana-leaf-wrapped ice cream" (name only); leaf containers for ice cream in Indonesia and India, which are containers, not wraps. Leaf cones such as clorot and cucurucho are not frozen.

• **Limits:** most Thai retail and delivery sites could not be read, and the OTOP registry has no public keyword search. Only Google Patents pages were readable (WIPO and the Thai DIP portal were not). The retail absence is weak evidence.

• **Verdict:** novel as a leaf-wrapped frozen *drink* on the evidence found. The Ratchaburi precedent is disclosed above. Full record: rework/25_novelty_search_extended.md.

4. Technical basis (virtual screen, lab reference)

Question	Result	Basis
Surface starts to melt at 30 C	0.22 h (about 13 min)	Transient radial heat model, h = 10 W/m2K
50% thawed (sippable)	1.53 h	Same model; grid difference 0.44% at t100 (8 vs 16 nodes)
Fully thawed	3.26 h	Lumped estimate was 2.47 h; conduction lags the core
Exposed-edge leak within 2 h	100% of 200 runs	Vessel-sized paths wick the wrap in 1.5 to 7 min
Sealed-edge critical damaged-path radius	0.24 um median (5 to 95%: 0.14 to 0.45 um)	Monte Carlo, 200 runs over 1 to 3 mPa s; freeze-damage size is unmeasured
Intact cuticle	does not leak	Pores 0.3 to 2 nm, far below the threshold
Continuous inner film	no liquid path	Not plastic-free

Thaw and wicking model: rework/17. Leak screen: rework/16. Leaf optics for banana now sourced (section 9). Not measured: freeze damage and seam strength.

5. Thai conditions: thaw timing and holding rules

Bangkok Mar-May normals (TMD 48455, 1991-2020): mean maximum 34.6 C in March and 35.7 C in April, humidity 72 to 74 %, dew point 23.9 to 24.9 C (April 24.8 C), 6.3 to 7.6 h of sun per day. The model uses a diurnal mean of 31.5 C with 4.3 K amplitude, a half-sine sun peak of 800 W/m2, and a gel-pack melting point of about -1.3 C. Splash and open-water temperature is set to 29.3 C, the Bangkok canal annual mean, as the nearest sourced proxy. Every input and its confidence is in rework/22_thailand_inputs.md. Virtual screen, not measured.

Scenario	Surface starts to melt	50% thawed	Core above 5 C	Max surface (C)
S0 Lab reference, 30 C still air	0.22 h	1.53 h	not in window	6.1
S1 Street stall, shade, hot-season day	0.11 h (5 to 95%: 0.09 to 0.14 h)	0.84 h	1.96 h	35.6
S2 Street stall, full sun, hot-season day	0.09 h (5 to 95%: 0.07 to 0.10 h)	0.63 h	1.49 h	49.5
S3 Parked car, closed, 2 h	0.12 h (5 to 95%: 0.11 to 0.13 h)	0.75 h	1.61 h	32.7
S4 Insulated box + gel packs, motorbike, 3 h	1.51 h (5 to 95%: 1.18 to 2.56 h)	not in window	not in window	0.2
S5 Extreme 40 C day, full sun, bare pop	0.06 h (5 to 95%: 0.05 to 0.07 h)	0.45 h	1.15 h	51.3
S6 Reefer at -18 C, 30 min door open	1.12 h	not in window	not in window	0.0
S7 Home freezer outage, 28 C kitchen	0.20 h	1.45 h	3.21 h	26.8
SU1 Festival stall in sun, hot afternoon, 3 h	0.08 h	0.57 h	1.37 h	46.8
SU2 Songkran street stall, splash-wetted, 2 h	0.02 h	0.24 h	0.80 h	30.4
SU3 Ice-water cooler at a picnic, 4 h	0.05 h	1.31 h	not in window	2.8
SU4 Insulated cooler with gel packs at a festival, 4 h	1.55 h	3.18 h	not in window	2.7
SU5 Parked car at a festival lot, 3 h	0.12 h	0.72 h	1.56 h	52.3

S0 is the lab reference (30 C still air, no sun). S1 to S7 are the earlier Thai exposures; SU1 to SU5 are summer occasions. Times are from leaving the freezer. Monte Carlo ranges are shown for S1 to S5 only. Figure 22 shows the traces.

• **Thai holding rules (food premises):** non-heat ready-to-eat food, beverages included, must be kept below 5 C, or may stay at room temperature for no more than 2 hours (2566 notification, clause 6(4)). No Thai time-out-of-freezer rule was found.

• **Frozen storage:** the Thai –18 C rule covers fresh food in premises. No Thai rule was found for processed or packaged frozen products, or for transport. The design target is –18 C, following the proposed Codex draft (final text not retrieved).

• **Notification No. 463 on edible ice:** in force from 12 December 2025. It requires ice to meet the drinking-water standard, and it requires the label "Ice for Consumption" on packaged ice. It has no drink-specific wording.

6. Summer occasions: do the conditions still hold?

Occasion	Drink-side threshold (um)	Outside water	Outside threshold (um)	Top of wrap likely wet
Lab reference, 30 C still air	0.180	none	n/a	no
S1 Street stall, shade, hot-season day	0.170	condensate	0.163	yes
S2 Street stall, full sun, hot-season day	0.168	condensate	0.244	yes
SU1 Festival stall in sun, hot afternoon, 3 h	0.167	condensate	0.257	yes
SU2 Songkran street stall, splash-wetted, 2 h	0.162	splash	0.131	yes
SU3 Ice-water cooler at a picnic, 4 h	0.165	ice water	0.131	yes
SU4 Insulated cooler with gel packs at a festival, 4 h	0.715	none	n/a	no
SU5 Parked car at a festival lot, 3 h	0.171	condensate	0.232	yes

Thresholds: the smallest damaged-path radius that wicks the 100 mm wrap in the time available. Paths at or above it leak, so higher means more margin. The outside-water column assumes only the bottom edge is wet, and it uses the measured 76 degree leaf contact angle. The drink-side value uses the lower-bound viscosity (1 mPa s), so it is the conservative end of the section 4 range. The damaged-path size is unmeasured, so these are thresholds, not leak probabilities.

• **Thaw window:** does not hold in full sun. Half the drink is sippable in about 0.6 h and the whole pop passes 5 C at about 1.5 h. In shade the two-hour window only just holds: the core passes 5 C at about 2.0 h.

• **Best hot-season option in the screen:** an insulated cooler with gel packs. The surface starts to melt at about 1.6 h, half the pop thaws at about 3.2 h, and the drink-side threshold rises to about 0.7 um.

• **Drink-side leak margin:** roughly holds. The threshold falls only from 0.180 um to 0.162 to 0.171 um, because the melt starts earlier.

• **Outside water:** not established. Dew on banana and plantain leaves lasted 14 plus or minus 2 hours a night on 16 of 18 nights at one humid-tropics site (Costa Rica), and the screen caps retained dew at 90 um per side (a model cap). Songkran splash and ice-water coolers wet the wrap as well. For all of these the top of the wrap is likely wet, so the screen gives no climb margin.

• **Contact angle, resolved by physics:** intact waxy cuticle (about 130 degrees for fresh leaf) gives no capillary climb. Damaged paths with exposed cell wall (76 degrees) do wick. The screen applies 76 degrees to the damaged paths only.

• **Verdict:** the leak claim does not hold for summer outdoor use by default. It holds only if the wrap stays dry above the melt line, the damaged paths are narrower than the values above, and the damaged paths are the only ones that wick, as the screen assumes (intact wax does not).

7. Quantum runs

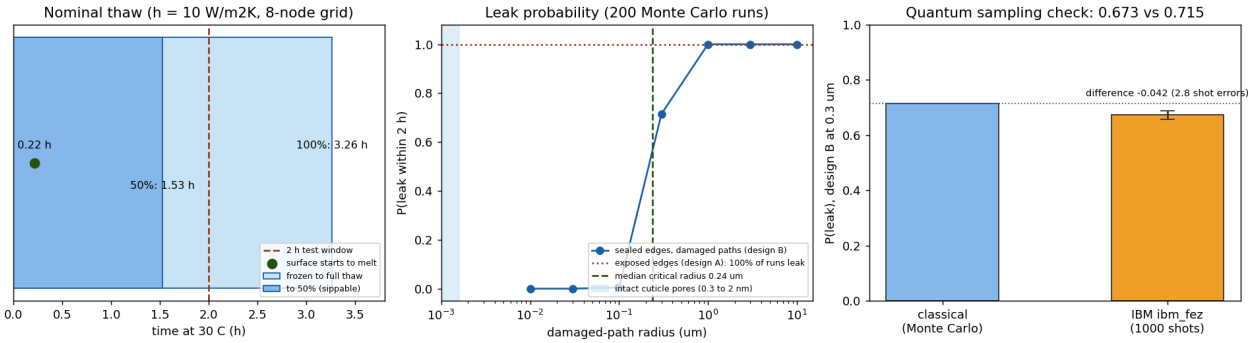
Run	Result	Note
One-qubit sampling, ibm_fez, 1000 shots	0.673 +/- 0.015	Classical value 0.715; 2.8 standard errors low; plausibly hardware noise
Amplitude estimation, exact simulation	0.7177 +/- 0.0016	local simulation
Amplitude estimation, fake_fez noise model	0.7122 +/- 0.0016	local simulation
Amplitude estimation, ibm_kingston hardware	0.5171 +/- 0.0017	job dba4kuc24mrc7384st0g

Run	Result	Note
Local sweep (no IBM), 6 Grover powers, 250 to 4000 shots, 30 repetitions	std 0.0028 classical vs 0.0007 noiseless at 36000 oracle calls	Noiseless error falls roughly as 1/sqrt(N), 3.3-fold over a 16-fold budget. The noisy model is biased by +0.0015 at 4000 shots, a small offset that does not match hardware (see figure 21).

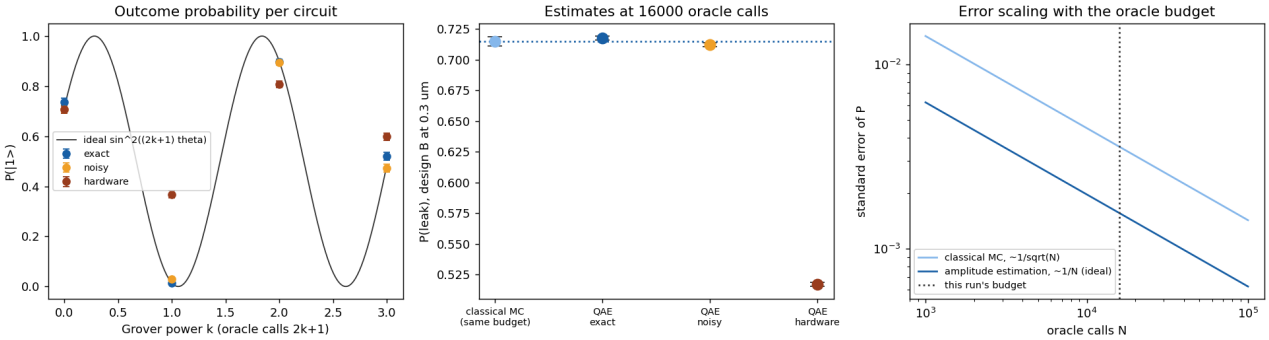
Reading: the quantum runs test the method, not the leak question. Hardware noise dominates the deeper circuits (ibm_kingston gave 0.517 against 0.715), and the local noise model does not reproduce that error, so no precision gain is claimed. The leak decisions do not depend on the quantum results.

8. Figures

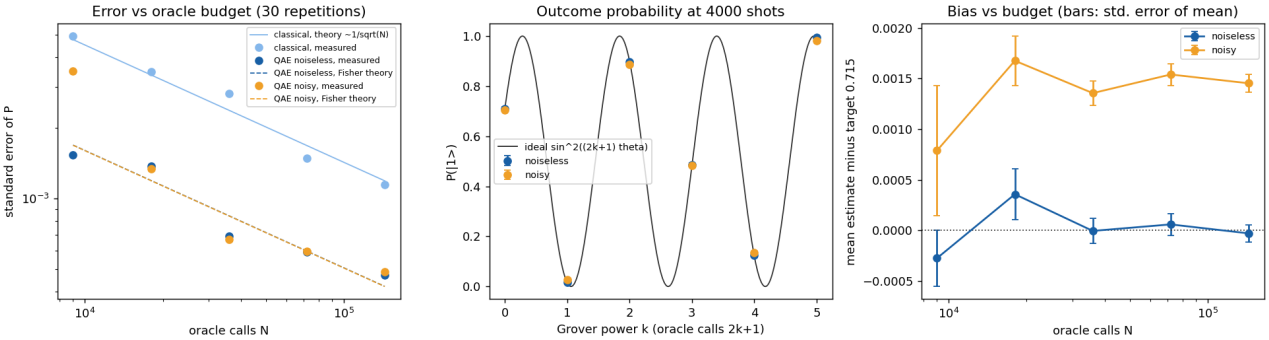
Leaf-wrapped frozen pop: thaw, leak and quantum-check results (virtual screen, not measurements)



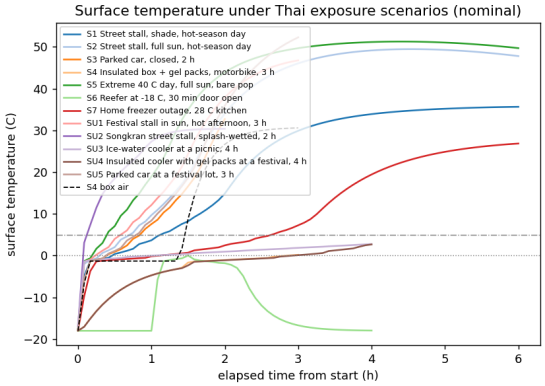
Amplitude estimation of the leak probability (virtual screen, real and simulated runs)

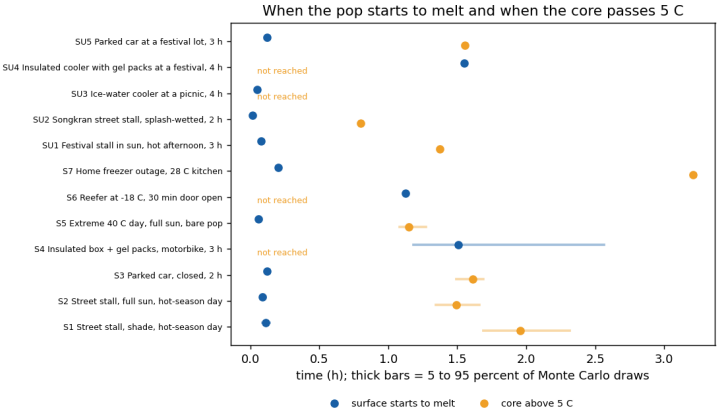


Local quantum simulation sweep, design B leak probability (no IBM hardware used)

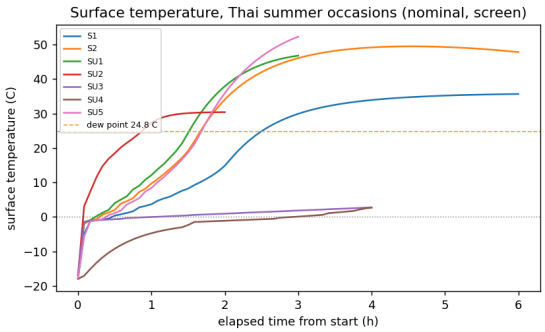


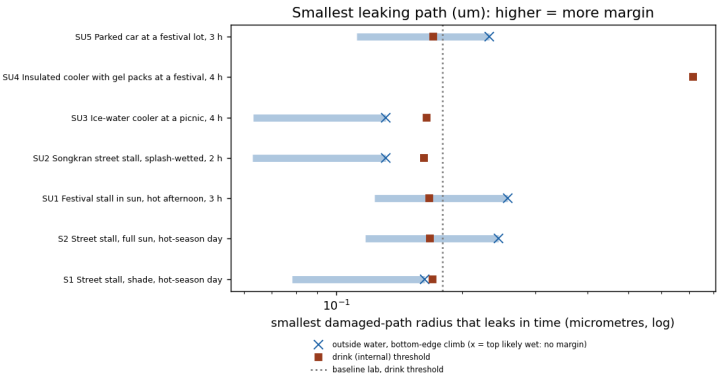
Thailand stress test, frozen leaf-wrapped pop (virtual screen, not measured)





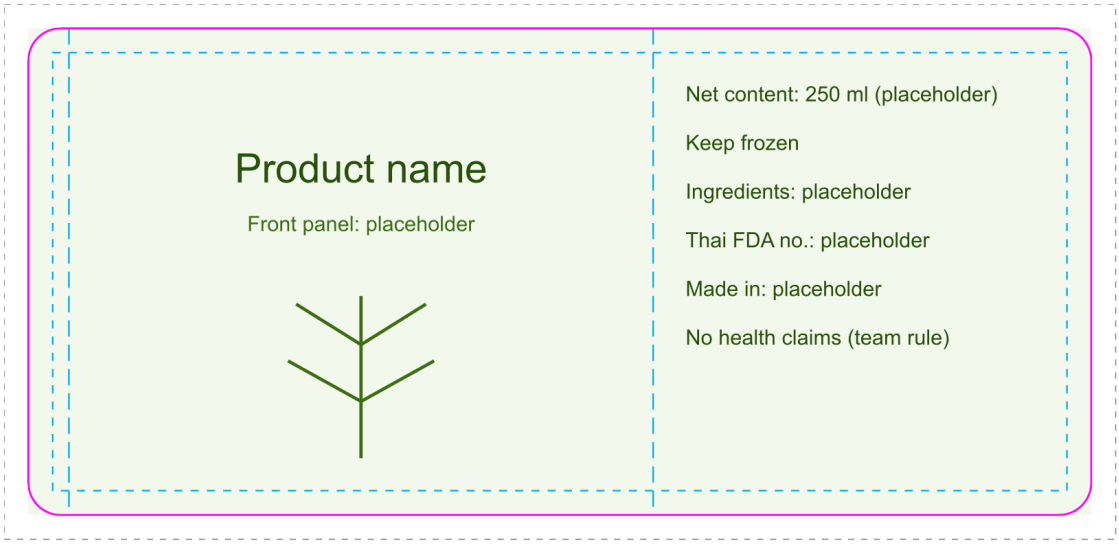
Thai summer occasions: thaw and wicking screen (virtual, damaged-path size unmeasured)





9. Product visuals and leaf optics





Working draft. Assumed: pop 40 mm across; label 131 x 60 mm finished, 3 mm bleed.

Measure a prototype before printing. Label placement on the leaf is untested.

All text is placeholder. Thai FDA label content needs a registration adviser.

Cut magenta, safe area blue, bleed grey. Hide the Guides and Bleed layers before printing.

Mockup: CC0 leaf, ice and paper textures from ambientCG. The leaf is a generic leaf photo, not confirmed as banana or pandan. The earlier approved still is rework/23; the leaf-optics version is rework/26. Banana leaf optics are applied since 18 October: colour scaled to the sourced 10.5 % green reflectance (Gausman and Allen 1973, derived), translucency weight 0.12 from the sourced 12 % transmittance. The mapping is derived and not validated. Dimensions are assumptions. Label text is placeholder. Thai FDA label content needs a registration adviser.

10. Open items: status

Item	Status	Decision or limit
Novelty: global, patents, OTOP, retail	Closed (limited)	Searched. One precedent disclosed. Retail, OTOP and WIPO limits stated in section 3.
Banana leaf optics	Closed (resolved)	Measured and derived values sourced; not applied to the locked render.
Pandan leaf optics and cuticle roughness	Closed (physics assumption)	Banana values used for pandan (plus or minus 20 %); roughness 0.3. The screens do not depend on them.
Splash and open-water temperature	Closed (proxy)	Bangkok canal annual mean, 29.3 C, used. No Songkran or tap value found.
Thai frozen storage rule for packaged products	Closed (physics assumption)	Target –18 C: most water is ice, so the thaw starts slow. Codex draft is the reference.

Item	Status	Decision or limit
Thai time out of freezer	Closed (physics assumption)	The thaw screen governs the temperature history; the Thai 2-hour rule is the legal limit for sale.
Condensate on packaged frozen food	Closed (physics assumption)	Mass transfer gives about 15 g/m2/min uncapped; the 90 um cap is reached in about 6 to 22 min. The screen uses the dew-point film rule.
Leaf contact angle (76 vs 130 degrees)	Closed (physics)	Intact wax does not wick; damaged paths at 76 degrees do. The outside-water rows apply to damaged paths.
Product classification (beverage or ice cream)	Working assumption	Treated as a non-heat ready-to-eat beverage, pending a Thai registration adviser. If classed as ice cream, the –2.2 C storage rule applies instead.
Physical leak test: leaf-only wrap (rework/33)	Confirmed by user, not yet run	Dye-penetration on 300 wraps, controls included. Pass if zero leaks. The lab run is a separate physical step and needs its own go-ahead. A simulated run is in rework/34.
Stomatal state in the leaf-only test (rework/34)	Adopted 18 Oct (rework/33, step 7a)	The simulation shows the outcome depends on open or closed stomata, not only on breach fraction. The scoring rule is adopted for any run. The lab run is not scheduled in the desk plan.
Second track: isoflavone soy drink (rework/40)	Closed as limit; not advanced	Desk track complete. The food-or-supplement route, the dose, the safety and the cost stay unverified. No outside contact is in the plan. The leaf wrap (A) remains primary.
PET pouch compliance (rework/32)	Confirmed by user	Thai No. 435 limits verified. Overall migration margin about 76x; antimony needs antimony-free PET; fat content and a migration test (needs approval) are open.
Unknown parameters (drink, damage, stomata, process)	Closed (physics)	Resolved in rework/30: water activity 0.995, pH 6.3, lag from h0 = 2. Leaf-only seal not supported; inner barrier is the working design.
Percolation of damage (rework/31)	Closed (physics)	Connected airspace network adopted. Leaf-only seal survives only below about 1e-8 damaged cells; 50 percent leak near 3 to 5 x 1e-7.
Taste of inner barrier (rework/31)	Closed (physics)	PE and PP scalp aroma completely; PET takes up about 0.2 percent in 2 h. PET selected; EVOH the alternative.
Consumer effect of inner barrier (rework/31)	Closed (decision)	Positioned on leaf format and peel-and-sip; the plastic-free buyer hypothesis is withdrawn for the barrier version. Consumer test out of scope.
Spoilage after thaw (rotten condition)	Closed (physics screen)	Growth screen: sealed drink spoils in about 6 to 12 h at 30 C; contaminated, in about 3.5 to 7 h. The 2-hour rule governs. Rules in rework/28.
Pathogens and microbial load	Closed (physics screen)	From one cell per serving, hazard in about 6 to 27 h at 30 C. The start load decides the risk, so a kill step or pathogen-free raw material is required. Rules in rework/29.

• Widen the global search later, if the team wants a full patent claim review (CN106414847A is the one pending candidate to read).

• Confirm the claim wording and classification with a Thai registration adviser, including the rules on packaging and claim wording.